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## 0.3-1 | Shear Stiffness Example

**Table 1.1**: Bearing Specifications

| variable | value | [value] | description |
|---|---|---|---|
| G_1 | 58.00 p_si | 0.40 MPA | shear modulus |
| K_1 | 200000.00 p_si | 1378.95 MPA | bulk modulus |
| rnl | 51 | 51 | number of rubber layers |
| rdia | 39.50 inch | 100.33 cm | bearing diameter |
| rthk | 0.40 inch | 1.02 cm | layer thickness |
| sdia | 38.00 inch | 96.52 cm | shim diameter |
| sthk | 0.09 inch | 0.23 cm | 11 guage shim thickness |
| cpi | 3.1418 | 3.1418 | contant pi |

**Bearing Stiffness for Circular Bearing**

**Eq. 1.1:** rubber height

```
rht = rnl·rthk

rht = 20.40 inch     [rht] = 51.82 cm   | rubber height
```

| rthk | rnl |
|---|---|
| 0.40 inch | 51 |
| layer thickness - | number of rubber layers |

**Eq. 1.2:** bearing height

```
bht = rht + sthk·(rnl - 1)

bht = 24.95 inch     [bht] = 63.37 cm   | bearing height
```

| sthk | rht | rnl |
|---|---|---|
| 0.09 inch | 20.40 inch | 51 |
| 11 guage shim thickness | rubber height - | number of rubber layers |

**Eq. 1.3:** shear stiffness

$$Ks_1 = \frac{G_1 \cdot cpi \cdot rdia^2}{4 \cdot rht}$$

$Ks_1$ = 3.48 k_in [$Ks_1$] = 6.10 kN_cm | shear stiffness

| rht | $G_1$ | rdia | cpi |
|---|---|---|---|
| 20.40 inch | 58.00 p_si | 39.50 inch | 3.1418 |
| rubber height | shear modulus | bearing diameter | contant pi |

**Shape Factor for Circular Bearing**

**Eq. 1.4:** shape factor 1

$$sh_{1\ 1} = \frac{sdia}{4 \cdot rthk}$$

$sh_{1\ 1}$ = 24 [$sh_{1\ 1}$] = 24 | shape factor 1

| sdia | rthk |
|---|---|
| 38 inch | 0 inch |
| shim diameter | layer thickness |

**Eq. 1.5:** shape factor 2

$$sh_{2\ 1} = \frac{bht}{rdia}$$

$sh_{2\ 1}$ = 1 [$sh_{2\ 1}$] = 1 | shape factor 2

| bht | rdia |
|---|---|
| 25 inch | 40 inch |
| bearing height | bearing diameter |

## 0.3-2 | Natural Frequency

The natural frequency (n_f) in Hz of a mounted body on a spring is

n_f = 5 / sqrt(x)

where x is the effective deflexion of the spring in centimetres. The effective deflexion is affected (and usually decreases) by the amplitude effect and by non-linear springs (Fig. 16).

[Figure: fig16 _static_load_deformation_curve.png] Fig. 16.–OAB is a ‘static’ load deformation curve. Its exact position and shape will depend on the amount of structure breakdown. The static stiffness at A is the slope of the tangent AC. When a dynamic, small amplitude deformation ED is superimposed on the static deformation at A its stiffness is the slope of AF. Thus the effective dynamic deformation FG is smaller than CG or OG, and the natural frequency will be higher than that predicted from static behaviour. C and F are generally closer to O than illustrated, especially with gum and lightly-filled natural rubber vulcanizates.

[Figure: fig17_transmissibility_graph.png] Fig. 17.–The transmissibility of a natural rubber containing 40 parts by weight of carbon black. Temperature 19 degrees C. Experimental data of Snowdon, J. C., Brit. J. Appl. Phys, 1958, 9, 461-9. The full line is the theoretical relationship for no damping.

Transmissibility

Transmissibility (T) is the ratio of the amplitude of the vibration on the ‘protected’ side of the spring to that of the disturbing vibration, or when the spring is on a rigid base, the ratio of the corresponding forces. T depends on the ratio of the disturbing frequency n to the natural frequency n_f of the mounted system (see above).

Under normal operating conditions when n > 3*n_f natural rubber gives attenuation comparable with materials having no damping (Fig. 17). At resonance, when n = n_f, the small amount of damping in natural rubber prevents the peak value of T becoming excessive.

Stiffness Characteristics

A mathematical theory of rubber-like elasticity has been developed for elastic deformations of up to several hundred per cent. The theory applies strictly only to ideal rubbers, which are completely reversible, incompressible and isotropic, but in practice actual rubber springs have been shown to conform quite well. The behaviour of certain kinds of rubber spring may therefore be calculated for deformations of any practicable amount.

However, when the deformation is complex, an exact solution may be unattainable. In some of these cases approximate relationships can be derived from the classical theory of elasticity. This theory, which is the basis of standard engineering practice, only applies to materials in which the strains are very small (a few per cent), but the errors introduced by extrapolation to strains of the order of 10-20% are not excessive. In the stiffness equations which follow, terms beyond the first have only been given where their contribution is likely to exceed the variability of the elastic constants.

Calculated stiffnesses should be within +-15% of the actual stiffness in most cases; small adjustments can be obtained by slight changes of rubber hardness.

In general, stiffness is particular to a given direction, the stiffnesses in other directions may sometimes be an order of magnitude different as, for example, in bridge bearings. By the suitable selection and location of two or more units the stiffness of a composite spring in three different directions can be varied independently. The design of such springs, of which the inclined shear mounting is a typical example, requires only a knowledge of mechanics and the principal stiffnesses of the units.

*Fig. 18.*
*Shear mounting.*

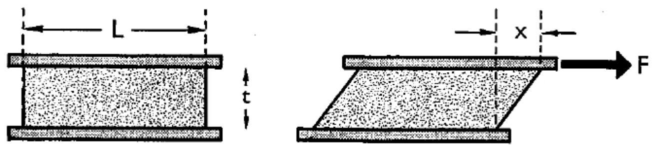


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**Fig. 2.1** - Shear Mounting